1 **(a)** Titanium dioxide, TiO_2, is a white solid, which is an amphoteric oxide. In the structure of titanium dioxide, the titanium ion is bonded to six oxide anions.

(i) Complete the electronic configuration of a titanium atom.

$1s^2$ ..... **[1]**

(ii) Suggest the shape around the titanium ion in titanium dioxide.

..... **[1]**

(b) **(i)** Aluminium oxide is another example of an amphoteric oxide.
Write two equations to illustrate the reaction of Al_2O_3 with an acid and a base of your choice respectively.

.....

..... **[2]**

(ii) The ionic radius of Al^{3+} is 0.050 nm and Ti^{2+} is 0.086 nm.
Explain the difference in ionic radii between Al^{3+} and Ti^{2+}.

.....

.....

.....

..... **[2]**

(c) Titanium(II) chloride is prepared by the thermal decomposition of $TiCl_3$ at 500°C. The reaction is driven by the loss of volatile $TiCl_4$.

$$2TiCl_3\ (s) \rightarrow TiCl_2\ (s) + TiCl_4\ (g)$$

(i) State and explain the sign for $\Delta S^{\ominus}$.

.....

..... **[1]**

(ii) Deduce the sign of the enthalpy change, ΔH, of the thermal decomposition of $TiCl_3$, given that the decomposition is spontaneous only at high temperature. Explain your answer.

.....

.....

..... **[1]**

(iii) Explain why $TiCl_3$ forms a violet solution, but $TiCl_4$ forms a colourless solution.

.....

.....

.....

..... **[2]**

(d) Another transition element that is bonded to oxygen atoms is manganese.

Two examples are manganate(VI) ion, MnO_4^{2-}, and manganate(VII) ion, MnO_4^-.

(i) Given that the structure of MnO_4^{2-} is similar to that of SO_4^{2-}, draw the 'dot–and–cross' diagram of MnO_4^{2-} and state its bond angle.

Bond angle: ..... **[2]**

(ii) Acidified potassium manganate(VII), $KMnO_4$, and acidified potassium dichromate, $K_2Cr_2O_7$, can be used as oxidising agents in organic reactions.

With reference to relevant $E^\ominus$ values, suggest why $KMnO_4$ is a stronger oxidising agent than $K_2Cr_2O_7$.

.....

.....

.....

..... **[2]**

(e) **A**, **B** and **C** are isomers with the molecular formula, $C_5H_{10}O$, that contains one or two of the following functional groups.

- Alkene
- Alcohol
- Carbonyl

Reactions are carried out on **A**, **B** and **C** and the observations are shown in Table 1.1.

Table 1.1

	with acidified $K_2Cr_2O_7$ (aq)	**with acidified $KMnO_4$ (aq)**	**with 2,4–DNPH**	**with Br_2 (aq)**
A	orange to green	purple to colourless	no reaction	no reaction
B	no reaction	no reaction	orange precipitate	no reaction
C	no reaction	purple to colourless	no reaction	orange to colourless

(i) **A** is a cyclic compound and does not rotate plane of polarised light. Draw the structure of **A**.

A

[1]

(ii) **B** is a symmetrical molecule. Draw the structure of **B**.

B

[1]

(iii) Write the equation for the reaction which occurs when **A** reacts completely with an excess of acidified potassium dichromate(VI). Use [O] to represent the oxidising agent in the reaction.

[1]

(iv) State all possible functional groups in **C**.

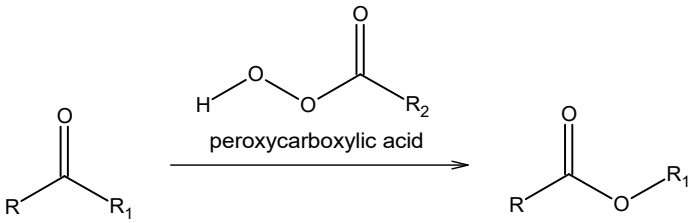
..... **[1]**

(v) **C** is unable to exhibit stereoisomerism. Draw the structure of **C**.

C

[1]

(f) Ketones can undergo oxidation forming esters through the Baeyer–Villiger oxidation reaction by using peroxycarboxylic acids as shown in the equation below.



The first step of the mechanism of the Baeyer–Villiger oxidation reaction involves the nucleophilic attack of the lone pair of electrons on the oxygen atom bonded to the hydrogen atom in the peroxycarboxylic acid to the carbonyl carbon in the ketone.

Draw the first step of the mechanism of the Baeyer–Villiger oxidation reaction. Show all relevant dipoles, curly arrows and the structure of the intermediate.

[2]

[Total: 21]